Cluster 612

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Subjects

Research fields: Computer science, Biology, Mathematics
The most common research fields of articles in this cluster. Read more.

Research subfields: Data mining, Algorithm, Pattern recognition
Common research subfields of articles in this cluster. Read more.

Key concepts: proteomics data, mass spectrometry data, data analysis, tandem mass spectrometry, Google Scholar
These key concepts are identified algorithmically using article titles and abstracts. Read more.

13.01% of all articles are AI-related
Articles are algorithmically labeled as AI-related or not. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to robotics
Articles are labeled as robotics-related or not using an ETO model. The percentage is based on articles in the cluster from the five year period ending in 2023 with English-language abstracts. Read more.

0.30% of English-language articles are related to computer vision
Articles are labeled as computer vision-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to natural language processing
Articles are labeled as NLP-related or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to AI safety
Articles are labeled as related to AI safety or not using an ETO model. The percentage is based on articles from the five year period ending in 2023 with English-language abstracts. Read more.

0.00% of English-language articles are related to cybersecurity

Vital signs

669 recent articles in the cluster
669 articles in this cluster were published in the five year period ending in 2023.

Average article is **13.12** years old
The average age of articles in this cluster. However, most Map of Science metrics are calculated using articles from the past five years only.

Growth rating: **63.65th** percentile
Over the three year period ending in 2023, this cluster grew faster than roughly 63.65% of other clusters.

Citation rating: **58.28th** percentile
A cluster's citation rating is equal to the average citation percentile for all the articles in the cluster. Citation percentiles are calculated relative to articles of the same publication year.In this case, the average article in this cluster is cited more often than roughly 58.28% of all other articles published in the same year.

96.41% of articles are in English

Not expected to grow extremely fast
As determined by ETO's growth prediction model. Read more.

Countries and languages

Top author countries

Each row lists the number of articles in this cluster from the last five years with at least one author from an organization in the specified country. Data available for 616 of 669 articles. 513 of these articles had at least one author from an organization in these ten countries.

Country	Articles (five year period ending in 2023)	Yearly citations per article (average)
United States	236	4.63
China	126	3.81
Germany	89	17.40
United Kingdom	60	26.39
Belgium	42	4.42
Canada	37	2.96
France	33	3.55
Russia	32	4.49
Australia	25	3.93
Netherlands	21	3.65

Top funding countries

Data available for 329 of 669 articles. 282 of these articles disclosed funding from one or more organizations in these ten countries.

Country	Articles (last five years)	Yearly citations (average)
United States	129	14.54
China	69	6.08
Belgium	41	4.23
United Kingdom	34	40.24
Germany	30	45.52
Canada	11	1.83
Brazil	11	1.98
Russia	9	2.73
France	8	6.12
Spain	7	3.24

Articles and sources

Core articles

Core articles are especially highly connected to other papers within the cluster.

Sequence-to-sequence translation from mass spectra to peptides with a transformer model
2023: bioRxiv (Cold Spring Harbor Laboratory). 24 citations.

Comprehensive evaluation of peptide de novo sequencing tools for monoclonal antibody assembly
2022: Briefings in bioinformatics. 38 citations.

De novo peptide sequencing with InstaNovo: Accurate, database-free peptide identification for large scale proteomics experiments
2023: bioRxiv (Cold Spring Harbor Laboratory). 21 citations.

De novo mass spectrometry peptide sequencing with a transformer model
2022: bioRxiv (Cold Spring Harbor Laboratory). 35 citations.

Full-Spectrum Prediction of Peptides Tandem Mass Spectra using Deep Neural Network
2020: Analytical chemistry. 63 citations.

Oktoberfest: Open-source spectral library generation and rescoring pipeline based on Prosit
2023: Proteomics. 26 citations.

Accurate de novo peptide sequencing using fully convolutional neural networks
2023: Nature communications. 34 citations.

Review articles

Review articles have between 100 and 1000 out-citations. These articles describe and systematize other contributors' research.

Introduction to Mass Spectrometry Data
2022: Computational biology. 0 citations.

Benchmarking the identification of a single degraded protein to explore optimal search strategies for ancient proteins
2023: bioRxiv (Cold Spring Harbor Laboratory). 3 citations.

Rescoring Peptide Spectrum Matches: Boosting Proteomics Performance by Integrating Peptide Property Predictors into Peptide Identification
2024: Molecular & cellular proteomics : MCP. 6 citations.

Validation Methods of Peptide Identification Results in Proteomics
2023: 生物化学与生物物理进展. 1 citations.

Proteomic repository data submission, dissemination, and reuse: key messages
2022: Expert review of proteomics. 10 citations.

Software Options for the Analysis of MS-Proteomic Data
2021: PROTEOMICS DATA ANALYSIS. 3 citations.

Algorithms for de-novo sequencing of peptides by tandem mass spectrometry: A review
2023: Analytica chimica acta. 14 citations.

Highly cited articles

Highly cited articles are the articles in the cluster with the most citations overall.

- The ProteomeXchange consortium at 10 years: 2023 update
  2022: Nucleic acids research. 332 citations.
- iProX in 2021: connecting proteomics data sharing with big data
  2021: Nucleic acids research. 384 citations.
- Proline: an efficient and user-friendly software suite for large-scale proteomics, 2020
  2020: England). 180 citations.
- Identification of modified peptides using localization-aware open search
  2020: Nature communications. 193 citations.
- The PRIDE database resources in 2022: a hub for mass spectrometry-based proteomics evidences
  2021: Nucleic acids research. 4049 citations.
- Bioinformatics Methods for Mass Spectrometry-Based Proteomics Data Analysis
  2020: International journal of molecular sciences. 198 citations.
- Proteome Discoverer—A Community Enhanced Data Processing Suite for Protein Informatics
  2021: Proteomes. 175 citations.

Top sources

The most common sources of articles from the last five years in this cluster, such as specific journals, conference proceedings, or preprint servers. Data available for 645 out of 669 articles. 343 of these articles came from these sources.

Source	Articles (last five years)
bioRxiv (Cold Spring Harbor Laboratory)	138
Journal of proteome research	83
Methods in molecular biology (Clifton, N.J.)	24
Proteomics	17
Bioinformatics (Oxford, England)	16
Analytical chemistry	16
Journal of the American Society for Mass Spectrometry	14
Research Square (Research Square)	13
Journal of proteomics	11
BMC bioinformatics	11

Authors, organizations, and funders

Top authors

The authors with the most articles from the last five years in this cluster. Data available for 667 of 669 articles. 113 of these articles were written by these authors.

Author	Articles (last five years)	Yearly citations (average)
Wout Bittremieux	25	15.88
Yasset Pérez-Riverol	25	105.84
Yasset Perez-Riverol	24	224.13
William Stafford Noble	24	10.86
Eric W. Deutsch	21	39.75
Fahad Saeed	20	1.41
Juan Antonio Vizcaíno	17	246.50
Mathias Wilhelm	15	17.42
Ralf Gabriels	15	19.10
Tim Van Den Bossche	14	19.44

Top author organizations

The author organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 621 of 669 articles. 153 had at least one author from one of these ten organizations.

Organization	Articles (last five years)	Yearly citations (average)
University of Washington - United States	37	7.87
European Bioinformatics Institute - United Kingdom	29	50.88
Institute for Systems Biology - United States	26	11.11
University of California, San Diego - United States	23	9.52
VIB-UGent Center for Medical Biotechnology - Belgium	21	4.72
Ghent University - Belgium	21	4.91
Florida International University - United States	20	0.69
Technical University of Munich - Germany	19	5.82
Chinese Academy of Sciences - China	18	1.63
University of Antwerp - Belgium	18	3.52

Organization types

This table covers the 596 articles in this cluster from the five year period ending in 2023 with data about their author organization types (out of 669 articles total). Read more

Commercial	Education	Nonprofit	Government
1.95%	33.28%	17.85%	3.50%

Top industry organizations

The industry organizations with the most articles in this cluster. Each row lists the number of articles in this cluster from the last five years with at least one author from the specified organization. Data available for 51 out of 669 articles. 27 of these articles had at least one author from an organization listed below.

Organization	Articles (last five years)	Yearly citations (average)
Thermo Fisher Scientific (United States)	9	6.12
Bioinformatics Solutions (Canada)	6	1.58
Thermo Fisher Scientific (Germany)	4	6.23
GALAB Laboratories (Germany)	3	2.33
TRIA Bioscience (United States)	3	2.89
InSysBio (Russia)	3	1.06
Dornier (Germany)	2	1.88
Eli Lilly (United States)	2	1
Bristol-Myers Squibb (Germany)	1	12.6
AstraZeneca (United States)	1	1

Top funders

Data available for 370 out of 669 recent articles. 209 of these articles disclosed funding from one or more of these organizations.

Funding organization	Articles (last five years)	Yearly citations (average)
National Institutes of Health - United States	86	20.22
National Science Foundation - United States	44	8.47
National Institute of General Medical Sciences - United States	39	7.41
National Natural Science Foundation of China	37	5.56
Ministry of Science and Technology - China	27	10.42
Biotechnology and Biological Sciences Research Council - United Kingdom	23	57.32
Federal Ministry of Education & Research (BMBF)	21	57.51
European Commission - Belgium	19	3.53
FWO	16	5.44
Wellcome Trust - United Kingdom	16	82.38

Funder types

This table covers the 319 articles in this cluster from the last five years with data about their funding organization types (out of 669 articles total).

Commercial	Education	Nonprofit	Government
1.41%	5.78%	15.00%	77.81%

Patents and industry

1579 patents cite articles in this cluster (**99.68th** percentile among all clusters)

16.42% of articles in the cluster are cited by patents

1.87% of articles have at least one author from industry

Top patent titles

Patent title	Same-cluster citations
PRECURSOR SELECTION USING AN ARTIFICIAL INTELLIGENCE ALGORITHM INCREASES PROTEOMIC SAMPLE COVERAGE AND REPRODUCIBILITY	34
SYSTEMS AND METHODS FOR DE NOVO PEPTIDE SEQUENCING FROM DATA-INDEPENDENT ACQUISITION USING DEEP LEARNING	26
METHODS AND SYSTEMS FOR DE NOVO PEPTIDE SEQUENCING USING DEEP LEARNING	26
METHODS AND SYSTEMS FOR ASSEMBLY OF PROTEIN SEQUENCES	20
METHOD FOR QUANTITATIVE ANALYSIS OF COMPLEX PROTEOMIC DATA	19
METHODS FOR DATA-DEPENDENT MASS SPECTROMETRY OF MIXED BIOMOLECULAR ANALYTES	18
CORRELATED PEPTIDES FOR QUANTITATIVE MASS SPECTROMETRY	14
METHODS FOR PEPTIDE MASS SPECTROMETRY FRAGMENTATION PREDICTION	13
PROTEOMIC IDENTIFICATION OF ANTIBODIES	13

Top patent assignees

Patent assignee	Same-cluster citations
UNIVERSITY OF WISCONSIN SYSTEM	53
WARTSILA OYJ	49
BIOINFORMATICS SOLUTIONS INC	42
COON JOSHUA J	39
JOHN DEERE (UNITED STATES)	36
WESTPHALL MICHAEL S	34
BAILEY DEREK	34
MCALISTER GRAEME	34
THEBOEING COMPANY	122

Connections

Top citing clusters

Cluster	Number of significant citations	CSET extracted phrases
35786	421	DIA data analysis, mass spectrometry, Data-Independent Acquisition Mass, DIA data, Google Scholar
47144	146	Proteomic Tumor Analysis, Clinical Proteomic Tumor, Proteomics, Tumor Analysis Consortium, Breast Cancer
43364	120	human gut microbiome, human gut, metaproteomics data analysis, microbial communities, gut microbiome metaproteomic
5201	113	metabolomics data analysis, mass spectrometry, Mass spectrometry-based metabolomics, untargeted metabolomics data, data analysis
6703	108	tandem mass tag, Quantitative proteomics, mass spectrometry, isobaric mass tags, TMT labeling
46200	100	single cells, single-cell proteomics, mass spectrometry, sample preparation, individual cells
34312	76	HLA class, human leukocyte antigen, HLA peptide presentation, cancer cells, mass spectrometry
50247	74	Top-down mass spectrometry, top-down proteomics, proteoforms, intact proteins, mass spectrometry analysis
13061	73	mass spectrometry, targeted proteomics, protein biomarkers, peptides, reaction monitoring mass
29898	63	Cross-Linking Mass Spectrometry, Protein structure, protein complexes, mass spectrometry analysis, Quantitative Protein Interaction

Top cited clusters

Cluster	Number of significant citations	CSET extracted phrases
35786	118	DIA data analysis, mass spectrometry, Data-Independent Acquisition Mass, DIA data, Google Scholar
6703	59	tandem mass tag, Quantitative proteomics, mass spectrometry, isobaric mass tags, TMT labeling
5201	33	metabolomics data analysis, mass spectrometry, Mass spectrometry-based metabolomics, untargeted metabolomics data, data analysis
47144	31	Proteomic Tumor Analysis, Clinical Proteomic Tumor, Proteomics, Tumor Analysis Consortium, Breast Cancer
13061	31	mass spectrometry, targeted proteomics, protein biomarkers, peptides, reaction monitoring mass
4254	31	PHOSPHOPEPTIDE ENRICHMENT, Protein phosphorylation, affinity chromatography, Google Scholar, biological samples
42224	25	Human Protein Atlas, HUPO Human Proteome, Uncharacterized Human Proteins, missing proteins, Proteome Project
34312	17	HLA class, human leukocyte antigen, HLA peptide presentation, cancer cells, mass spectrometry
50247	15	Top-down mass spectrometry, top-down proteomics, proteoforms, intact proteins, mass spectrometry analysis
9366	15	N-linked Intact Glycopeptides, N-linked glycopeptide enrichment, intact glycopeptide analysis, mass spectrometry, protein glycosylation

Top GitHub repositories

Repository	Number of within-cluster citations	Number of stars
smdb21/java-miape-api	5	0
tzom/ms2pip_c	5	0
xiaoping-yang/ms2pip_c	5	0
paretje/ms2pip_c	5	0
compomics/ms2pip	5	38
pwilmart/precursor_mass_corrections	5	0
MaryamMary/MetaMorpheus	4	0
cullanm/MetaMorpheus	4	0
ianhirsch/MetaMorpheus	4	0
nf-core/mhcquant	4	33